



GREENLIGHT ECOLOGY LTD

Bat Survey Report

Serge hill Lane – Lunch Room and New Shed

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GREENLIGHT ECOLOGY LTD

Contents

1 EXECUTIVE SUMMARY 2

2 INTRODUCTION..... 2

3 LEGISLATION 3

4 PROPOSED DEVELOPMENT 3

5 SCOPE OF WORKS 3

6 METHODOLOGY 4

7 Desk Study 4

8 PRELIMINARY ROOST ASSESSMENT 4

9 Hibernation Assessment 5

10 Bat Emergence Re-entry Survey (BERS) 5

11 Assessment Summary 5

12 CONCLUSIONS..... 6

13 RECOMMENDATIONS 6

14 REFERENCES 6

1 EXECUTIVE SUMMARY

- 1.1.1 Greenlight Ecology Ltd were engaged by Studio Benjamin (The Client) to undertake a Preliminary Roost Assessment (PRA) to support proposed plans in advance of potential change of use of an existing barn and creation of a shed at Featherbed Ln, Abbots Langley WD5 ORZ (the site).
- 1.1.2 A full exterior and interior inspection and hibernation assessment of the building was undertaken and found no evidence of bats having used the building, and the building was unlikely to be used for hibernating bats.
- 1.1.3 The potential for roosting bats has been assessed as **low** based on the methodology developed by Collins (2023) and therefore one bat emergence and re-entry survey (BERS) was recommended.
- 1.1.4 The BERS was carried out on 21st July 2025 with the first bat activity recorded at 21:28 and no emergences.

2 INTRODUCTION

2.1 Project Background

- 2.1.1 Greenlight Ecology Ltd were engaged by Studio Benjamin (The Client) to undertake a Preliminary Roost Assessment (PRA) to support proposed plans of potential change of use of an existing barn and creation of a shed at Featherbed Ln, Abbots Langley WD5 ORZ (the site). The OS Grid Reference at the approximate center of site is TL 10766 04254.
- 2.1.2 A full Preliminary Roost Assessment was scoped following the current methodology described in Bat Surveys for Professional Ecologists Collins (V4 2023).
- 2.1.3 A site habitat map and location plan is provided as **Appendix 1** of this report.

2.2 Limitations and Disclaimers

- 2.2.1 This report takes no account of seasonal variation or fluctuations in the presence of any protected species that might take up residence following this report. Lack of signs of a particular species does not confirm absence, merely that there was no indication of its presence at the time of survey. This survey does not constitute a full site assessment for Protected Species (only bats) or invasive species such as Japanese Knotweed. A search for the presence of Tree Preservation Orders (TPOs) on site has not been undertaken. Should the proposed development not be underway within 24 months of the date of issue of this report, it is strongly recommended that the assessment be repeated.
- 2.2.2 Greenlight Ecology Ltd accept no responsibility of whatever nature to third parties to whom this report, or any part thereof, is made known. Any such party relies on the report wholly at its own risk.
- 2.2.3 The client may not assign the benefit of the report or any part to any third party without the written consent of Greenlight Ecology Ltd.
- 2.2.4 Ownership of copyright of all electronic and printed versions or parts of this report (including appendices) remains with Greenlight Ecology Ltd.

3 LEGISLATION

3.1.1 All British bat species receive legal protection in the United Kingdom. The Wildlife and Countryside Act 1981 (WCA) (as amended) transposes into UK law the Convention on the Conservation of European Wildlife and Natural Habitats (Bern Convention). The 1981 Act was recently amended by the Countryside and Rights of Way (CROW) Act 2000 and the more recent Habitats Regulations amendments (2017). All British bat species are listed under Schedule 5 of the 1981 Act, and is therefore subject to the provisions of Section 9, which makes it an offence to:

- Intentionally kill, injure or take a bat [Section 9(1)];
- Possess or control any live or dead specimen or anything derived from a bat [Section 9(2)];
- Intentionally or recklessly disturb a bat while it is occupying a structure or place which it uses for shelter or protection [Section 9(4)(b)];
- Intentionally or recklessly obstruct access to any structure or place which a bat uses for shelter or protection [Section 9(4)(c)]; and
- Sell, offer for sale, possess or transport for the purpose of sale or publish advertisements to buy or sell a bat [section 9(5)].

3.1.2 Bats are also included on Annex IV of Council Directive 92/43/EEC of 21 May 1992 on the Conservation of Natural Habitats and of Wild Fauna and Flora (known as the Habitats Directive). As a result of the UK ratifying this directive, all British bats are protected under The Conservation Regulations 1994 (the Habitat Regulations) (as amended 2007). Annex IV of the Habitats Directive requires member states to construct a system of protection as outlined in Article 12, this is done through Schedule 2 of the Regulations whereby Regulation 39 makes it an offence to:

- Deliberately capture or kill a bat [Regulation 39(1)(a)];
- Deliberately disturb a bat in such a way as to be likely to significantly affect i) the ability of any significant group of that species to survive, breed, rear, or nurture their young OR ii) the local distribution of that species [Regulation 39(1)(b)]; and
- Damage or destroy a breeding site or resting place of a bat [Regulation 39(1)(d)].

3.1.3 Under the law, a roost is any structure or place used for shelter or protection. This could be any structure e.g., any building or mature tree. Bats use many roost sites and feeding areas throughout the year. These vary according to bat age, condition, sex, and species as well as seasonality and weather. Since bats tend to reuse the same roosts for generations, the roost is protected whether the bats are present or not.

4 PROPOSED DEVELOPMENT

4.1.1 It is understood that the proposed works are for the change of use for an existing corrugated metal barn and the construction of a new Resi- Ancillary shed.

5 SCOPE OF WORKS

5.1.1 The scope of works consisted of the following;

- Undertake a PRA as detailed in Bat Surveys for Professional Ecologists: Good Practice Guidelines (Collins 2023) with a full external and where possible, internal inspection of the building associated with the proposals regarding bats;
- Identify any physical evidence of bat activity;
- Determine the building's potential to support bats; and
- Indicate emergence surveys as/if required to determine likely presence/absence.

6 METHODOLOGY

- 6.1.1 The assessment work summarised in this report was undertaken by James Sweetman of Niche Ecology Ltd (NE Class license number 2021-51575-CLS-CLS), an experienced and trained bat surveyor with 19 years' experience.
- 6.1.2 A desk study was undertaken, with bat records for locations within 2 km of the site purchased from Herts Environmental Records Centre (HERC).
- 6.1.3 The PRA was undertaken on site as detailed in section 8.

7 Desk Study

- 7.1.1 EPSL data - A search of the magic.gov.uk database for granted EPSLs within a 2km radius of the site has been completed alongside an intensive desk study and record centre analysis. No EPSLs are present within a 2km radius of the site, and no records of bat activity have been documented in the last 5 years.

8 PRELIMINARY ROOST ASSESSMENT

- 8.1.1 The PRA and hibernation assessment was undertaken on the existing corrugated shed on 21st of July 2025 between 3pm and 5pm.
- 8.1.2 The building was fully accessible to the surveyor for the inspection.
- 8.1.3 The assessment was undertaken to identify any bats and/or their signs to determine the likelihood of the building supporting them. This information was used to assess the degree to which bats use the buildings, if at all, and subsequently whether emergence survey(s) would be required.
- 8.1.4 The building's exterior was examined from ground level using binoculars and a high-powered torch to look for gaps or other suitable features that bats may use, as well as bats themselves and evidence of use such as droppings and/or feeding remains (e.g., moth wings).
- 8.1.5 The single-story barn roof is pitched at approximately 45 degrees and constructed from corrugated metal sheeting. Corrugated roofing sheets were laid directly onto the wooden roofing timbers, with no roofing membrane or insulation present. Potential does exist for crevice-dwelling bat species to roost between the wooden beams/joists and corrugated metal (sheltering on top of the beams during the day). The roof ridge is formed by overlapping metal sheets; daylight is visible along the entire length of the roof ridge, providing access into the barn for bats.
- 8.1.6 Externally, corrugated sheeting material has been fixed to the wooden structure of the barn. A narrow internal void likely exists between the external metal cladding and internal timber planks and plyboard, creating suitable daytime roosting conditions for bats. Given the probable fluctuating temperature extremes, the cavity spaces are less likely to be used as a maternity roost but could be used by individual bats.
- 8.1.7 The interior of the building was inspected by the surveyor for features as well as bats themselves and evidence of use such as droppings and/or feeding remains (e.g., moth wings). Presence of cobwebs often indicates lack of use, and no evidence of bat activity was found.
- 8.1.8 Internally, rough sawn softwood roofing timbers are evident. The internal walls are constructed of horizontal timber planks, and plywood sheeting has been attached to the timbers. The barn is divided into three separate rooms. The mid-section is double height and vaulted to the ceiling/roof. Two mezzanine levels are located at each end of the barn, directly over each of the enclosed room areas. The mezzanine levels are notably darker and more sheltered than the ground floor of the barn. Sections of plyboard and MDF board (attached to the underside of the pitched ceiling) created a sheltered void that could provide potential roosting opportunities for bats during the daytime. The ground floor of the barn was relatively well lit due to the presence of windows, reducing its suitability as a day-roost. Gaps under the eaves (above the door frame) provide suitable access into the central double-height barn area for bats. However, a thorough search of each of the three separate ground floor internal areas revealed no presence of bat habitation. No bat droppings

were found on any of the floor areas, furniture, stored machinery, walls or window ledges. Areas below roof/ceiling beams were searched for droppings and feeding remains, but none was found.

8.1.9 The building was subsequently classified by the potential to support bats using the categories developed by the Bat Conservation Trust. The survey was carried out in accordance with Bat Surveys for Professional Ecologists: Good Practice Guidelines (Collins 2023).

8.1.10 No other suitable habitats or features were present on site.

8.1.11 See appendix 2 for photographs taken during survey.

9 Hibernation Assessment

9.1.1 As part of the PRA, an assessment of the building was undertaken to assess if the building would be suitable for hibernating bats.

9.1.2 Given the probable fluctuating temperature extremes of corrugated metal construction, the cavity spaces are unlikely to support hibernating bats.

9.1.3 Therefore the hibernation potential of the building has been assessed as Negligible. No features present within the structure that are likely to be used by bats during the winter period.

10 Bat Emergence Re-entry Survey (BERS)

10.1.1 As the PRA survey assessed the barn as having low potential to support individual roosting bats one BERS was conducted on 21st July 2025. See Appendix 3 for proformas and full survey records.

10.1.2 The survey was carried out from 20:51 – 22:36 with sunset being at 21:06. The weather conditions were optimum see table below

Temp start	Temp End	Wind start	Wind end	Cloud start	Cloud end	Rain
17	16	1	1	3	3	1

10.1.3 The survey was lead by James Sweetman (NE Class license number 2021-51575-CLS-CLS) and assisted by Samuel Radonich Bsc MRes with over 5 years of experience bat surveying.

10.1.4 The first bat activity recorded was a soprano pipistrelle recorded foraging at 21:38. The last activity recorded was a common pipistrelle commuting at 22:35. Soprano and common pipistrelle were the only two species recorded during the survey.

10.1.5 No emergences were recorded during the survey.

11 Assessment Summary

11.1.1 The building is constructed of corrugated metal sheeting attached to soft wood timber frame.

11.1.2 The interiors were fully accessible to the surveyor who found no evidence of bat use such as recent droppings, feeding remains, or urine splashes. Furthermore, there cobwebs in the loft further indicating a lack of use by bats

- 11.1.3 The hibernation potential of the building has also been assessed as Negligible.
- 11.1.4 Based on the surveyor's inspection, and assessment, the building has been classified as having low roost suitability using the methodology developed by Collins (2023). As such one further survey was recommended.
- 11.1.5 The BERS conducted on 21st July 2025 recorded no emergences.
- 11.1.6 No other suitable habitats to support roosting bats were identified on site.

12 CONCLUSIONS

- 12.1.1 A PRA has been completed on the barn at Featherbed Ln, Abbots Langley WD5 ORZ on 21st July 2025. The assessment was undertaken in accordance with methodology stipulated in Bat Surveys for Professional Ecologists: Good Practice Guidelines (Collins 2023).
- 12.1.2 There was no evidence to suggest that bats use the building, and there are limited access points in the structure, and therefore the building has been classified as having low roost suitability for individual bats using the methodology developed by Collins (2023).
- 12.1.3 The barn was subjected to one BERS in accordance with methodology stipulated in Bat Surveys for Professional Ecologists: Good Practice Guidelines (Collins 2023).

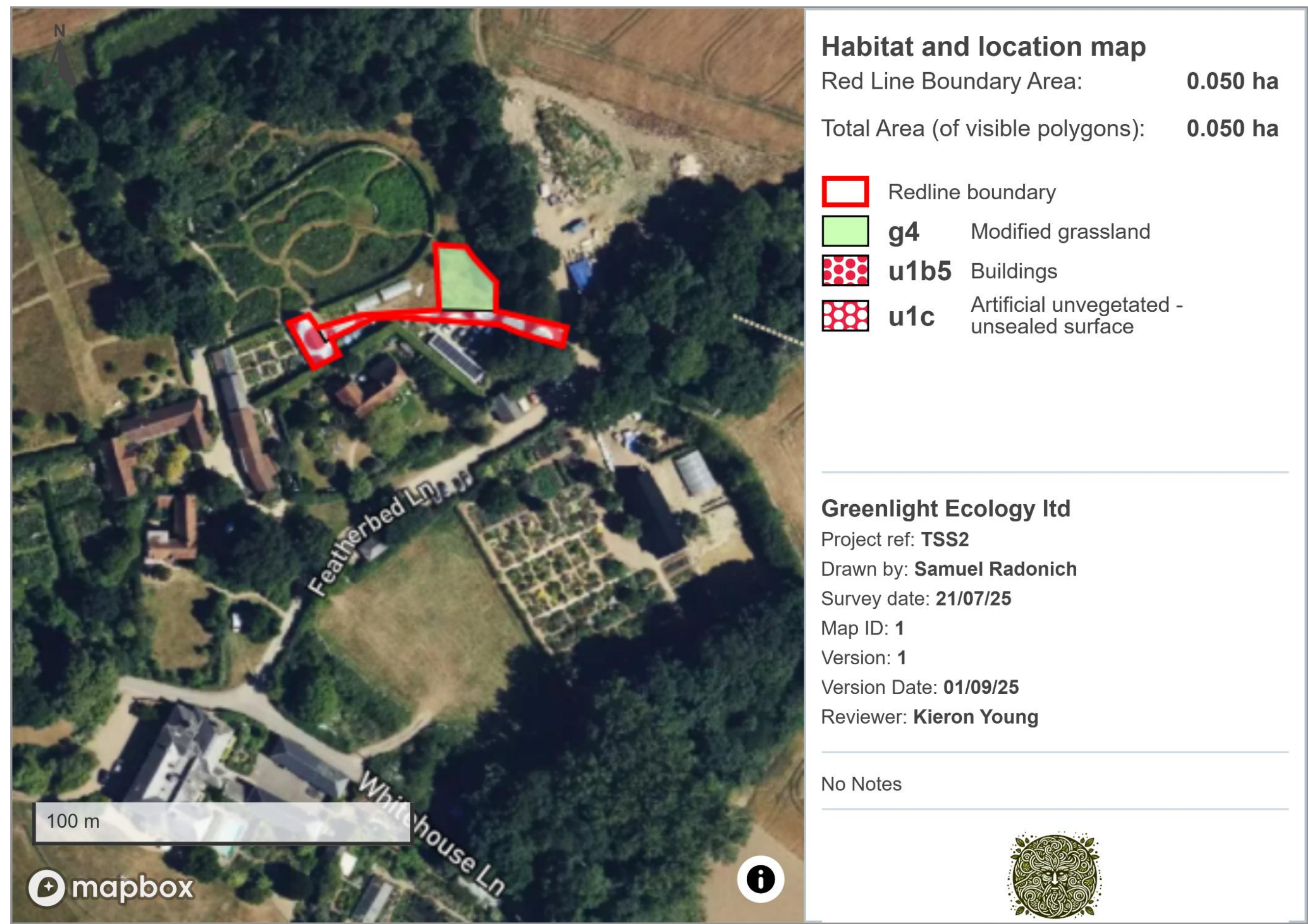
13 RECOMMENDATIONS

- 13.1.1 If bats are unexpectedly discovered during building works, then works should stop immediately and the advice of a licensed bat consultant sought.

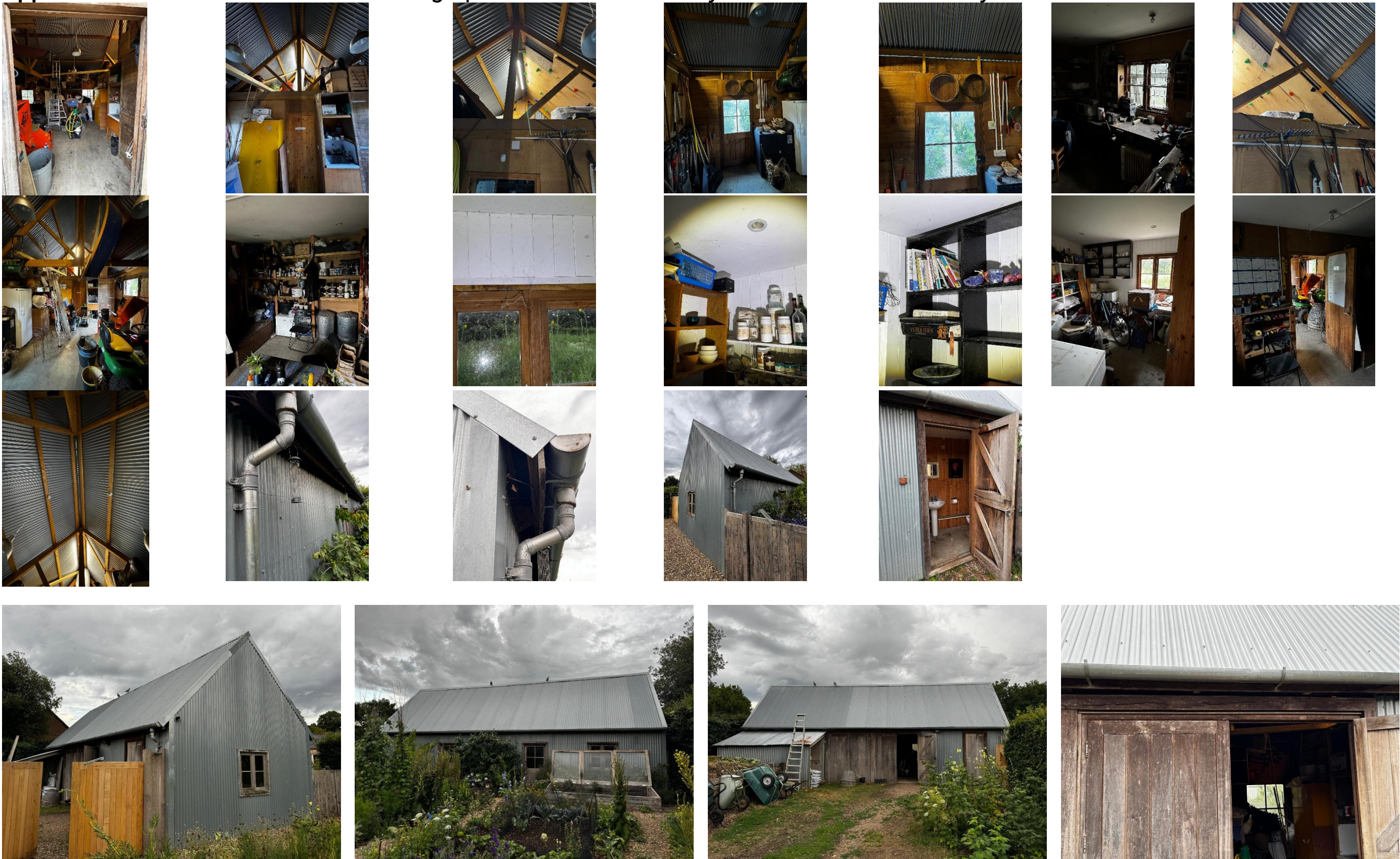
14 REFERENCES

- BCT (2023) Bat Surveys for Professional Ecologists: Good Practice Guidelines Ed; Collins, J (4th edition); BS 42020:2013 Biodiversity. Code of practice for planning and development;
- Greenlight Ecology Ltd Serge Hill – Lunch Room and New Shed PEA 2025
- HMSO The Conservation of Habitats and Species Regulations 2010; HMSO
- The Countryside and Rights of Way Act 2000;
- HMSO Wildlife and Countryside Act 1981;
- Mitchell-Jones A.J (2004) Bat Mitigation Guidelines. English Nature;
- Mitchell-Jones A.J. and McLeish A.P. (2004) The Bat Workers Manual 3rd Edition. Joint Nature Conservation Committee.

Appendix 1 – Habitat and Location Map

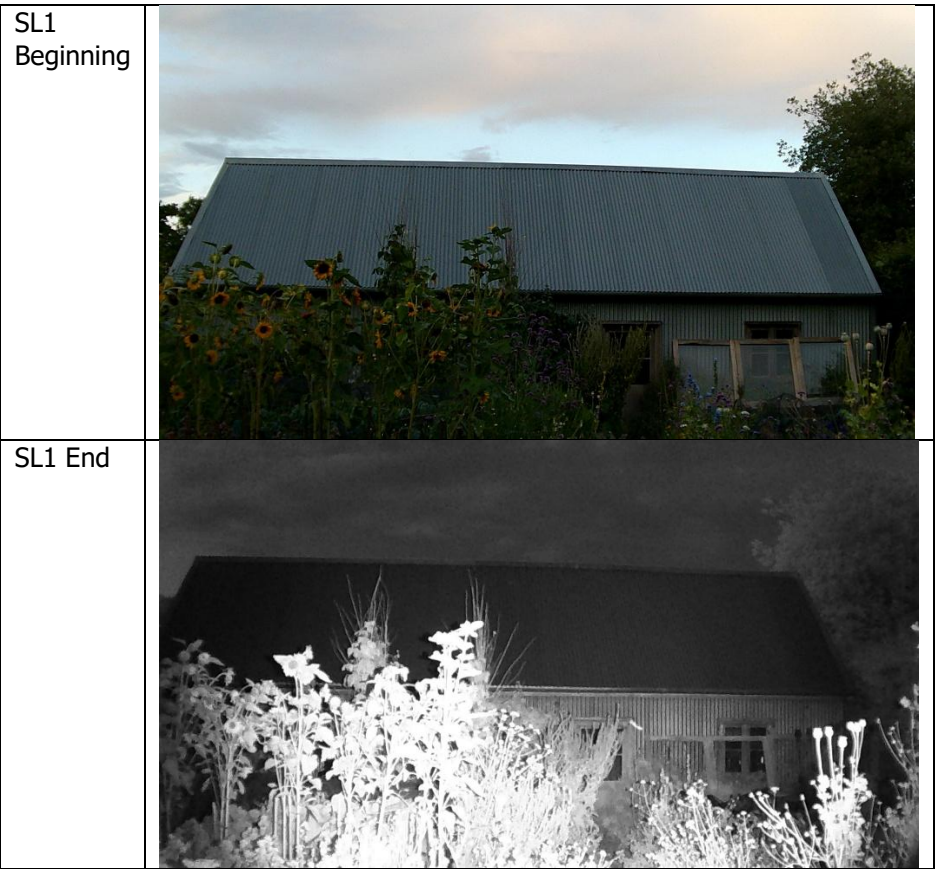


Appendix 2 – Internal and External Photographs from the Preliminary Roost Assessment Survey



Appendix 3 – BERS Proformas and photo evidence

Date	Survey time	Sunset	Temp start	Temp End	Wind start	Wind end	Cloud start	Cloud end	Rain	
21/17/2025	8:51 - 10:36	21:06	17	16	1	1	3	3		1
Surveyor full name	Company	Bat detector	Licence	Years bat survey experience						
Samuel Radonich	Greenlight Ecology	EMT pro 2	No	5						
Time	Species	Heard/seen	behaviour	Notes	Review needed					
21:38	Sp	H	F	Single pass						
21:07	Cp	H	F							
22:27	Cp	H	F							
22:29	Cp	H	F							
22:34	Cp	H	F							
22:35	Cp	H	F	Multiple passes						



Date	Survey time	Sunset	Temp start	Temp End	Wind start	Wind end	Cloud start	Cloud end	Rain	
21/17/2025	8:51 - 10:36	21:06	17	16	1	1	3	3		1
Surveyor full name	Company	Bat detector	Licence	Years bat survey experience						
James Sweetman	Niche Ecology	EMT pro 2	No	19						
Time	Species	Heard/seen	behaviour	Notes	Review needed					
22:03	Cp	H	F	Multiple passes						
22:04	Cp	H	F	2 bats single pass						
22:09	Sp	H	C	1 pass						
22:12	Cp	H	C	1 pass						
22:24	Cp	H	C	1 pass						
22:35	Cp	H	C	1 pass						

